

Remote Friction Control on 3-dimensional Object Made of Polystyrene Foam Using Airborne Ultrasound Focus

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ABSTRACT

In this study, we propose a method to remotely change the friction of a three-dimensional surface made of polystyrene foam. In a previous study, a system that projects an image onto a two-dimensional polystyrene foam surface and irradiates ultrasound according to a finger's position in contact with the surface to change the friction has been proposed. In this study, we extended it to a three-dimensional shape and quantitatively evaluated the change in friction. It was confirmed that the frictional change could be obtained even when extended to three dimensions but that the frictional change was less likely to occur depending on how the polystyrene foam was fabricated. This technology can be applied for adding the tactile feel to a simple 3D mockup model made of polystyrene foam.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; *Touch screens*.

KEYWORDS

Tactile display, Surface haptics, Ultrasound

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1 INTRODUCTION

One of the artificially presenting tactile sensation methods is to represent texture by changing the friction on the surface. This research field is called "Surface haptics," which is often used to change the tactile feel of the touch panel surface. This type of haptic display can be classified into two types according to the technique. One is to use static electricity to attract the skin to the target surface and increase the friction, such as [2]. The other is to vibrate the target surface with ultrasound and reduce friction by using the squeeze effect such as [7]. The difference is that the former increases friction while the latter decreases it. By generating such friction changes according to the position of the fingertip, it is possible to generate a texture that matches the image on the screen when implemented on a touch panel.

For the latter type of friction-reducing display, a method called "ReFriction" has been proposed, in which friction is reduced by irradiating ultrasound remotely to oscillate the polystyrene foam surface [1]. An ultrasound phased array was used to create a focal point of high sound pressure to oscillate the target surface. In general, it is difficult to oscillate a hard surface remotely because ultrasound is reflected. However, it is possible to oscillate an acoustically soft surface such as polystyrene foam. As a result, they supposed that the squeeze effect was induced by remote excitation similar to existing ultrasound friction-reduction systems.

In the previous study, two advantages of this method were introduced: first, the contact surface can be made disposable since there is no need to install a device such as an oscillator on the contacting object. The other is that the frictional change can be generated on a free surface since the remote oscillation is considered to be independent of the macroscopic shape of the object, depending only on the material and the structure that changes the resonance state. In the previous study, as the first step to reducing friction remotely, the target was limited to a two-dimensional surface, and the contact position between the fingertip and the surface was detected by an infrared sensor that can measure in two dimensions.

In this study, we develop a prototype device that can control friction on a three-dimensional surface and evaluate its basic performance in order to verify the feasibility of using it on a free-form surface, which was mentioned in the previous study. A depth sensor is used to detect contact between the fingertip and the target

surface, and ultrasound is emitted to the contact point to reduce friction. We measure the vertical and tangential forces when the fingertip traces on a free-form surface to confirm whether the friction was reduced actually or not. We show that friction reduction is possible even on a three-dimensional surface. Suppose such friction reduction on a free-form surface can be realized. In that case, it will be possible to create a simple mockup using polystyrene foam and present the surface texture using the surrounding ultrasonic array. This would be a useful method for prototyping with tactile feedback of the texture.

The followings are the two main contributions of this paper. (1) We have confirmed that friction reduction occurs even on surfaces with three-dimensional shapes and have shown the possibility of extending the ReFriction system to three-dimensional objects. (2) We found that the friction of a three-dimensional surface made of polystyrene foam was reduced depending on the fabrication method.

2 RELATED WORK

2.1 Surface haptics

In recent years, research on tactile displays that change the surface friction during touch has attracted attention as a research field called "Surface Haptics." With the spread of smartphones, people are increasingly operating their devices by touching a flat surface that does not change haptically, thus a method to change the friction on such a surface enriches experiences and improves the usability. There are two main ways to realize Surface Haptics. One is to increase the friction, and the other is to decrease it. The friction-increasing type of tactile display mainly uses electrostatic force to increase the friction force between the finger or sheet and the surface of the object using the attraction force between them [2] [3] [4] [5]. On the other hand, a friction-reducing tactile display uses ultrasound to vibrate a solid surface, and the resulting squeeze effect and intermittent contact between the finger and the object reduce the friction when stroking on the surface [6] [7] [8] [9] [10]. By changing the friction to correspond to the image on the screen, it is possible to create a variety of textures and virtual uneven shapes.

In the previous study [1], a new method of friction reduction using ultrasound was proposed, which uses an ultrasound phased array to remotely oscillate the contact surface and reduce friction. Although there is a limitation in the material that the friction can be reduced only on the surface of polystyrene foam, it has been shown that the tactile sensation can be changed according to the projected image on a two-dimensional surface. The previous study have indicated that there are two advantages of this approach. The first is that the target surface can be made disposable, and the second is that it can be applied to any free three-dimensional surface. In this study, we focus on the latter point and verify whether the same friction reduction can be achieved on a three-dimensional object surface, which has not been investigated in the previous study.

2.2 Airborne ultrasound tactile display (AUTD)

In this study, we use an ultrasound phased array called airborne ultrasound tactile display (AUTD) to oscillate a polystyrene foam surface from a distance. The ultrasound phased array has been

proposed by Iwamoto et al. in 2008 [11] as a method to present invisible tactile sensations remotely in the air, and since then, various applications have been proposed in the field of HCI such as [12]. Since it can present various spatio-temporal tactile patterns, it is used to clarify the sense of touch [13] [14]. The following review paper [15] describes the research trend in this area in detail. Many applications other than the presentation of tactile sensation have been proposed, such as projecting images onto small, lightweight beads floating in the air [16] [17] [18], levitating an object larger than the wavelength [19], and controlling the behavior of lightweight objects such as balloons [20] and a ping-pong ball [21] by irradiating ultrasound.

3 EXPERIMENTAL SETUP

Fig. 1 shows the implemented setup, where AUTD is placed at an angle of 45 degrees, referring to the previous study. This angle was adopted based on the results of preliminary experiments in the previous study, which showed that people easily perceive frictional changes when the angle between the ultrasonic transducer array surface and the target surface was between 30 and 70 degrees. The 3D surface to be targeted in this study is also prepared on the condition that it is basically within the range of angles.

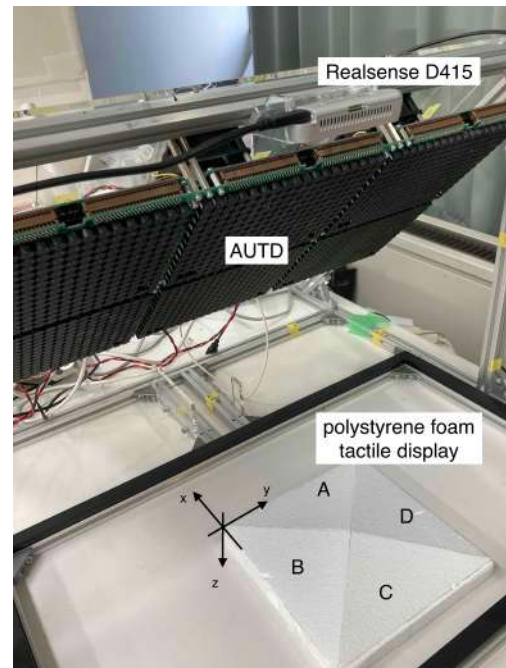


Figure 1: Implemented setup of our system

The difference from ReFriction system is the contact detection method. To detect the contact position of a fingertip, the previous study have used infrared sensors to detect the passage of a finger in a defined plane, but in this study, we used the RealSense D415 (Intel Corporation) to use depth maps and RGB information. For the finger tracking, we attached a blue sticker to the fingernail of the index finger and estimated its position in a two-dimensional plane by color tracking. For the surface of the object, we used the

information from the depth map and irradiated ultrasound when the fingertip was judged to be in contact with the surface. The detailed algorithm of the contact judgment is shown below.

- (1) Measure the initial depth information on a three-dimensional surface made of polystyrene foam from the top with RealSense. The surface is not moved after that, and contact with this depth information is judged.
- (2) Attach a blue sticker to the fingernail of the index finger, and color-track its 2D position. Based on this 2D information, the 3D coordinate $\mathbf{x}_f$ of the fingernail is obtained by using the depth information.
- (3) Calculate the 3D coordinate $\mathbf{x}_o$ of the point on the target surface corresponding to the 2D position of the fingertip.
- (4) When the distance between the 3D position $\mathbf{x}_f$ of the fingernail and the point $\mathbf{x}_o$ on the 3D surface directly below it is less than 40 mm, the ultrasound is focused on $\mathbf{x}_o$.

In general, the thickness between the fingernail and the fingertip is approximately 10 mm, but in this study, we set a larger thickness of 40 mm. This means that the ultrasound is irradiated even before the contact between the finger and the target surface. However, since the proposed method irradiates ultrasound on the target surface, the tactile sensation is not felt until the contact is made even if the stimulus is presented before the contact.

We prepared two types of target surfaces: a pyramidal surface (Fig.2) and a surface that varies sinusoidally in one direction (Fig. 5). The pyramidal surface has a side length of 200 mm and a height of 15 mm, and thus the tilt angle θ satisfies $\tan \theta = \frac{3}{20}$. Since $\arctan(\frac{3}{20}) = 8.5$ degrees, the angle between the surface and AUTD surface is approximately 45 ± 8.5 degrees, which is within the range of 30 to 70 degrees the preliminary experiment showed good sensitivity. During this fabrication process, the cut surface by the heat-type polystyrene foam cutter tended to show less frictional change. This is thought to be due to the change in properties to ultrasound caused by thermal denaturation during cutting (details will be discussed later). Since the surface sold as a product (mechanical cutting, or the surface itself at the time of foam molding) was more susceptible to frictional changes, the surface at the time of purchase was fabricated to become the surface of the target object.

Because of the limitations of the cutting method described above, the sinusoidal surface was cut out as a 3-5 mm thin, easily deformable polystyrene foam board, leaving the surface as it was when purchased, and bending and gluing it along sinusoidally deformed bases. The base was also made of polystyrene foam, but not all of the base was under the sinusoidal surface. The surface was supported at three lines, and some places in the surface were thin (3-5 mm).

4 EXPERIMENTS AND RESULTS

4.1 Quantitative evaluation on pyramidal structure

In order to confirm whether friction reduction occurs or not, a precise force measurement device (Tactile Force Plate, Tech Gihan Co., Ltd., TF-2020) was placed under the pyramid-shaped object as shown in Fig. 2, and the vertical and horizontal forces at that time were measured respectively. The pyramid has four faces, and



Figure 2: Force measurement device under the object

we call the one near AUTD face A, the far side C, the left side B, and the right side D, respectively as shown in Fig. 1. One of the authors traces each surface with the right index finger with the same amount of force, and measures the force in the three axial directions at that time. To be more precise, a method such as applying a constant pressing force with a robot arm and tracing the surface with an artificial finger could be considered. In this study, however, we considered that measurement by a human finger would be sufficient. This is because we can simultaneously measure both the vertical and horizontal forces. Thus we can estimate the friction coefficient even if it varies due to changes in the vertical force.

The direction of tracing was fixed to the left and right (y-direction), i.e., tracing with no change in height for the plane A and C and tracing with a change in height for the plane B and D. The normal and tangential force to the surface are calculated as follows.

$$\text{Normal force} = F_z \cos \theta \pm F_y \sin \theta, \text{ Friction} = F_y, (\text{A and C}) \quad (1)$$

$$\text{Normal force} = F_z \cos \theta \mp F_y \sin \theta, \text{ Friction} = F_y \cos \theta \pm F_z \sin \theta, (\text{B and D}) \quad (2)$$

Here, θ is the angle of the pyramid which satisfies $\tan \theta = \frac{3}{20}$.

Fig.3 shows a plot of the actual measured forces. The figure shows the vertical and horizontal force values with and without ultrasonic irradiation. It is clear from the figure that the frictional force F_y (blue line) decreases when the ultrasound is irradiated.

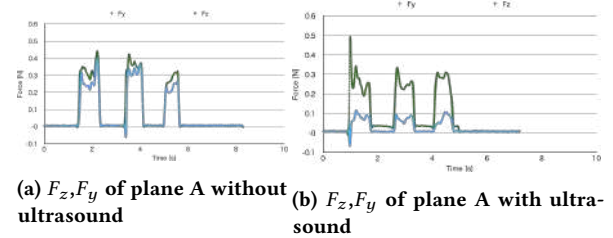


Figure 3: Time series F_z, F_y data of plane A.

Fig. 4 shows a two-dimensional plot of the results. The horizontal axis shows the normal force to the surface calculated by equations (1) and (2), and the vertical axis shows the friction force. Since the friction force is expressed as the product of the friction coefficient μ and the vertical force N , the slope of the line corresponds to the friction coefficient. Note that under the condition with ultrasound, the results tended to be separated only when tracing the surface C. Since one of the two distributions of pink plots is similar to the slope in the condition without ultrasound (green), it is thought that the two results were separated because the ultrasound was not

working continuously when tracing the C surface, and the ultrasound was switching between on and off states. For this condition, the regression line was drawn manually, and thus, it is not accurate one.

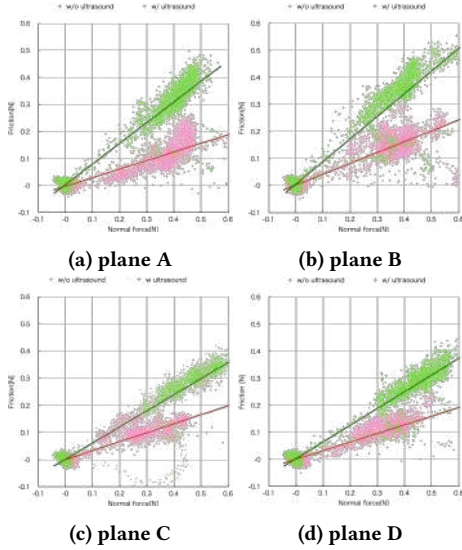


Figure 4: 2D plot of friction and normal force for each plane

Table 1: Slope of 2D plot graph with and without ultrasound

	A	B	C	D
With ultrasound	0.32	0.40	0.33	0.32
Without ultrasound	0.76	0.84	0.59	0.61

It is clear from Fig. 4 that ultrasound irradiation decreases the slope and the friction coefficient. The slope values for each surface are shown in Table 1. The values of the slope without ultrasound when tracing A and B planes (more than 0.75) are different from those when tracing the C and D sides (approximately 0.6). This may be due to the different ease of tracing for each surface orientation, as he used his right index finger for tracing. In all cases, the inclination was reduced to less than half of that without ultrasound, indicating that the friction was sufficiently reduced within the range of slope angles prepared in this study.

4.2 Qualitative evaluation on sinusoidal surface

We qualitatively confirmed whether frictional changes occur when ultrasound is irradiated while stroking a sinusoidally fluctuating surface. The spatial frequency of the sine wave shape was 30 cm, and the amplitude was 1.5 cm (peak-to-peak: 3cm). The direction of the sinusoidal variation was arranged parallel to AUTDs so that the change in the angle to AUTD surface would be small. One of the authors traced it with the index finger to see if the friction sensation changed at that time.

It is desirable to be able to quantitatively evaluate this surface based on a force sensor as well as the pyramidal surface. However, since this is the surface with a continuously changing inclination, it is difficult to accurately calculate the vertical and horizontal forces on the surface, as in the case of a square pyramid. Therefore, only a qualitative evaluation of whether the friction reduction is felt is conducted.

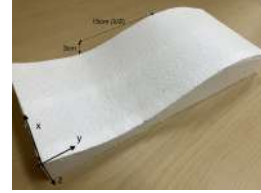


Figure 5: Tactile display with sinusoidal fluctuating surface

As a result, he could feel the reduction of friction, but the degree of friction change seemed to vary depending on the location. There are two possible reasons for this. One is the effect of the surface shape, and the other is the effect of the thickness of the polystyrene foam surface. In the present study, the sinusoidal surface was supported with three bases made of polystyrene foam, and the thickness of the other parts was thin polystyrene foam plates. In other words, the thickness of the part of the surface that was stroked was approximately 3 - 5 mm, depending on the location.

As mentioned above, it has been confirmed that the way polystyrene foam is processed has an effect on the tactile sensation. Taking this into consideration, we used a method of forming curved surfaces by deforming thin polystyrene foam sheets, but the effect of this process might be reflected in the results. More detailed verification should be done after a more stable forming method is developed. Since the slope changes continuously on a sinusoidal surface, it would be difficult to calculate the friction coefficient from the measured force of the mechanical sensor installed on the floor. Therefore, no further quantitative evaluation of this sinusoidal free-form surface will be conducted in this paper. However, we were able to qualitatively confirm that frictional changes occur not only on the combination of flat surfaces such as pyramid structure, but also on smoothly fluctuating surfaces.

5 DISCUSSION AND LIMITATIONS

By tracking and following the fingertip and irradiating focused ultrasound directly below the fingertip, we could confirm that a remote friction reduction effect occurred even on a three-dimensional surface. On the other hand, the friction on the surface C, which is far from AUTD, shows the bipolar result that about half of the cases with ultrasound show the same level of friction as the case without ultrasound. This result was attributed to the fact that there were many moments when the friction was not reduced due to the lack of stable ultrasonic irradiation under the finger. The reason for this is not apparent at the moment, but it may be due to errors such as color tracking markers not being detected because of reflections from different finger angles. Further investigation is needed.

As mentioned in Experimental Setup section, it was confirmed that the friction reduction was weakened on the surface cut by

the heated polystyrene foam cutter. We believe that this is due to the hardening of the cut surface caused by thermal denaturation during cutting. In the previous study, they assumed that the reason why the friction reduction was felt only on the polystyrene foam surface is that the surface is easily oscillated due to its low acoustic impedance. As a result, large surface deformation is excited and the surface friction is considered to be reduced due to the squeeze effect. If this understanding is correct, it is thought that the hardening of the surface due to thermal denaturation raises the acoustic impedance, making it harder to deform, and thus making it harder to reduce friction. It is necessary to consider the appropriate processing method to create a 3D model having an appropriately vibrating surface.

In this study, the sinusoidal surface was created by cutting a thin piece of polystyrene foam and deforming it in the one-dimensional direction while leaving the original surface intact. In this case, however, it was confirmed that the qualitatively perceived friction reduction effect was weakened in the unsupported thin part. The reason for this is not clear, however, it may be due to the thinness or the specific resonant structure of the surface. More detailed investigation is needed in the future.

The pyramid-shaped target surface used in this study reflects the results of preliminary experiments, and a less steep angle is used to keep within the range of angles where people can easily feel the friction change. In order to achieve this on a more free surface, it is necessary to arrange AUTD at multiple angles so that the ultrasound can always irradiate the target area from an appropriate angle. In this study, we presented the stimuli based on the initial depth information with the target surface fixed, however, we need to consider a more flexible method to generate friction reduction on a polystyrene foam model held in the hand. In such a case, how to detect the contact with the target will be a major issue, and will remain as a future research topic.

6 CONCLUSION

In this paper, the applicability of a method for reducing friction on a polystyrene foam surface by remotely applied ultrasound to a three-dimensional structure is examined. Since polystyrene foam is a material that can be easily shaped, prototyping including tactile sensation may be possible by creating a 3D mock-up model and adding tactile sensation to the surface.

In this paper, we prepared a pyramidal surface and a sinusoidally fluctuating surface, and evaluated each of them quantitatively and qualitatively. Quantitatively, we confirmed that the friction of the four surfaces of the pyramid-shaped object decreased by measuring the vertical and horizontal forces. It was confirmed that the friction coefficient decreased by about half from 0.6 to 0.3 in one typical case. It was confirmed that the friction change was felt on the sine fluctuating surface as well. On the other hand, it was suggested that the friction reduction effect may be suppressed depending on the structure of the surface. In addition, it was qualitatively confirmed that the effect also depended on the cutting method of the polystyrene foam, indicating that it will be necessary to

investigate a fabrication method suitable for such remote friction reduction in the future.

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